SHUBHAM CHATTERJEE

Process Engineer

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CONTACT

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QUICK FACTS

Experience 12+ Years
Current Shell
Notice 3 Months
CTC Confidential
Expected Negotiable

KEY SKILLS

- Process Simulation
- Relief & Flare Design
- Heat Exchangers (HTRI)
- Flow Assurance (OLGA, PIPESIM)
- HAZOP & LOPA
- Pump Hydraulics
- Brownfield MOC
- API , Shell DEPS
- Digital Twin & Digitalization
- AI/ML Predictive Analytics
- Data-Driven Problem Solving

SOFTWARE

- Aspen HYSYS
- UniSim Design
- FLARENET
- OLGA / PIPESIM/ Korf
- HTRI Xchanger
- Python / VBA
- SeeQ, Power BI
- Digital Twin Platforms, AI/ML Predictive Analytics

EDUCATION

B.Tech — Chemical Engineering
DIATM / WBUT
2009 – 2013
CGPA: 8.27 / 10.0
First Class w/ Distinction

LANGUAGES

English
Hindi
Bengali

CAREER SNAPSHOT

12+	Shell	3 Mo.
Years Experience	Current Employer	Notice Period

PROFILE SUMMARY

Process Engineer with 12+ years of experience spanning EPC detailed engineering and global operations support for LNG, GTL, and gas processing facilities. Proven ability to direct multidisciplinary engineering teams, govern complex scopes of work, and drive end-to-end technical delivery. Deep expertise in process simulation (HYSYS / UniSim), relief & flare systems (API 520/521), HTRI heat exchanger design, and hydraulic analysis, complemented by 8.6 years of plant-facing operations support at Shell. Adept at leading RCAs, HAZOP reviews, brownfield MOC, and continuous performance optimisation against API and Shell DEP standards. Actively driving digitalization initiatives, leveraging digital twin and AI/machine learning-based predictive analytics, to strengthen data-driven problem solving and asset performance.

PROFESSIONAL EXPERIENCE

Shell ▪ PTM Engineer (Process) — 2022 – Present

Bengaluru, India

- Developed Scope of work (SoW), managed deliverables, schedules, budgets, and multidisciplinary resources aligned with CTR/ToR requirements across global asset teams.
- Supported RCA investigations across global assets, identifying process and equipment reliability improvements that reduced recurring operational issues and improved facility availability.
- Supported Management of Change (MOC) for brownfield modifications ensuring technical integrity and process safety compliance.
- Defined and governed process control and safeguarding strategies aligned with P&ID philosophy, HAZOP/LOPA recommendations, and API 521, reducing risk exposure on critical facilities.
- Performed steady-state hydraulic simulations using OLGA supporting liquid management improvements.
- Led digitalization of process surveillance by driving adoption of digital twin models for critical systems, integrating real-time plant data to enable virtual performance monitoring and what-if scenario testing.
- Applied AI/machine learning-based predictive analytics to forecast equipment degradation and process deviations across global assets, enabling proactive interventions and enhanced problem-solving ahead of failure.
- Applied advanced troubleshooting methodologies combining first-principles process engineering with machine-learning-assisted trend analysis for faster problem identification.

Shell ▪ Senior Process Data Engineer/ Process Data Engineer — 2018 – 2022

Chennai, India

- Built deep operational expertise across gas gathering, separation, compression, and utility systems at multiple global Shell assets serving as the primary technical reference for process dynamics and troubleshooting.
- Architected and deployed comprehensive technical monitoring plans in a web-based surveillance platform, enabling proactive identification of performance drift on critical equipment and systems.
- Collaborated with asset engineers and operations teams to establish performance baselines and structured troubleshooting frameworks.

Petrofac Engineering Services ▪ Process Engineer 2014 – 2018

▸ *Chennai, India*

- Delivered detailed process engineering across offshore and refinery projects (FEED and EPC): P&ID/PFD development, H&MB, equipment sizing, datasheets, line lists, and hydraulic calculations.
- Performed relief valve and blowdown sizing; developed and validated flare network models using FLARENET per API 520/521 for complex multi-train facilities.
- Designed shell & tube heat exchangers using HTRI/TEMA; sized control valves, pumps, and vessels for hydrocarbon and utility services.
- Led and participated in HAZOP reviews, 3D model reviews, and process safety studies; developed control philosophy and DCS safeguarding strategies.

Key Projects

Greater Stella Area Development (North Sea) ▪ Client: Ithaca Energy

- Supported detailed engineering activities for the refurbishment and upgrade of a Floating Production Storage and Offloading (FPSO) facility in the North Sea. Delivered process engineering support through hydraulic studies, control valve sizing, process datasheet development, and process documentation updates, while contributing to pressure relief system evaluations and technical design reviews to ensure safe and reliable offshore operations.

Clean Fuel Project (Kuwait) ▪ Client: Kuwait National Petroleum Company

- Contributed to one of the world's largest refinery modernization and integration programs, supporting brownfield revamp activities across multiple refinery assets. Developed key process engineering deliverables, including process flow schemes, design documentation, pump hydraulics calculations and existing pumps and tanks adequacy check, and design change documentation, while performing pump hydraulic assessments and adequacy studies to support safe and efficient refinery operations.

Yibal Khuff Development Project (Oman) ▪ Client: Petroleum Development Oman (PDO)

- Provided process engineering support for the integrated development of gas processing, oil processing, sulphur recovery, and utility facilities within the Yibal Khuff field development project. Developed flare network models, performed relief valve sizing for multiple operating and emergency scenarios, executed control valve hydraulic analysis and depressurization studies using UNISIM, and supported process safety evaluations and system optimization initiatives across the facility.

EDUCATION

Bachelor of Technology (B.Tech.) — Chemical Engineering

Institution: Durgapur Institute of Advanced Technology and Management (DIATM)

University: West Bengal University of Technology (WBUT)

Year: 2009 – 2013

CGPA: 8.27 / 10.0 (First Class with Distinction)

CERTIFICATIONS & TRAINING

- Advanced Flare and Relief Systems — Network design, relief load evaluation, dynamic depressurisation
- Flow Assurance & Multiphase Flow (PFAM 2023) — Hydraulics, transient behaviour, system stability

- Introduction to Hydrates — Formation mechanisms, prevention, and mitigation
- Natural Gas Processing: LNG and NGL — Treatment, liquefaction, fractionation, specifications
- Masterclass: Improving Base Layer Control Sustainably
- Masterclass: COMPAS — Structured troubleshooting and performance monitoring

PERSONAL DETAILS

Nationality: Indian

Gender: Male

Marital Status: Married

Date of Birth: 28-Mar-1991